

§7.2 - Trigonometric Integrals

Definition #1

The functions tangent and secant are given by

$$\tan(x) = \frac{\sin(x)}{\cos(x)} \quad \text{and} \quad \sec(x) = \frac{1}{\cos(x)}$$

for $x \neq k\pi + \pi/2$ where $k \in \mathbf{Z}$.

Theorem #1

We have the Pythagorean identities

$$\cos^2(x) + \sin^2(x) = 1 \quad \text{and} \quad \tan^2(x) = \sec^2(x) - 1,$$

the power reduction/half-angle identities

$$\sin^2(x) = \frac{1 - \cos(2x)}{2} \quad \text{and} \quad \cos^2(x) = \frac{1 + \cos(2x)}{2}$$

and

$$\sin(x) \cos(x) = \frac{1}{2} \sin(2x).$$

Theorem #2

We have

$$\frac{d}{dx} \sin(x) = \cos(x) \quad \text{and} \quad \frac{d}{dx} \cos(x) = -\sin(x).$$

Also

$$\int \sin(x) \, dx = -\cos(x) + C \quad \text{and} \quad \int \cos(x) \, dx = \sin(x) + C$$

where $C \in \mathbf{R}$.

Theorem #3

We have

$$\frac{d}{dx} \tan(x) = \sec^2(x) \quad \text{and} \quad \frac{d}{dx} \sec(x) = \sec(x) \cdot \tan(x).$$

Also

$$\int \tan(x) \, dx = \ln |\sec(x)| + C$$

and

$$\int \sec(x) \, dx = \ln |\sec(x) + \tan(x)| + C$$

where $C \in \mathbf{R}$.

Example #1

Find

$$\int \cos^2(x) \, dx.$$

Trick:

$$\cos^2(x) = \frac{1 + \cos(2x)}{2}$$

then

$$\begin{aligned}\int \cos^2(x) dx &= \int \frac{1 + \cos(2x)}{2} dx \\ &= \int \frac{1}{2} + \frac{1}{2}\cos(2x) dx \\ &= \frac{x}{2} + \frac{1}{4}\sin(2x) + C\end{aligned}$$

where

$$\int \frac{1}{2}\cos(2x) dx = \int \frac{1}{4}\cos(u) du$$

is handled by a u-sub

$$u = 2x, \quad du = 2 dx, \quad \text{and} \quad dx = \frac{1}{2} du.$$

Example #2

Find

$$\int \cos^5(x) \, dx.$$

Trick:

$$\cos^2(x) = 1 - \sin^2(x)$$

Notice

$$\begin{aligned}\cos^5(x) &= (\cos^2(x))^2 \cos(x) \\ &= (1 - \sin^2(x))^2 \cos(x).\end{aligned}$$

$$\int \cos^5(x) dx = \int (1 - \sin^2(x))^2 \cos(x) dx$$

Apply a u -sub

$$u = \sin(x) \quad du = \cos(x) dx$$

then

$$\begin{aligned}\int (1 - \sin^2(x))^2 \cos(x) dx &= \int (1 - u^2)^2 du \\ &= \int 1 - 2u^2 + u^4 du \\ &= u - \frac{2}{3}u^3 + \frac{1}{5}u^5 + C.\end{aligned}$$

Back substitute to get

$$\int \cos^5(x) dx = \sin(x) - \frac{2}{3}\sin^3(x) + \frac{1}{5}\sin^5(x) + C.$$

Example #3

Find

$$\int \sin^3(x) \cos^5(x) \, dx.$$

Use Pythagorean identity,

$$\sin^2(x) = 1 - \cos^2(x)$$

and refactor our integrand

$$\begin{aligned}\sin^3(x) \cos^5(x) &= \sin^2(x) \cos^5(x) \sin(x) \\ &= (1 - \cos^2(x)) \cos^5(x) \sin(x) \\ &= (\cos^5(x) - \cos^7(x)) \sin(x).\end{aligned}$$

Let

$$u = \cos(x) \quad du = -\sin(x) dx$$

then

$$\begin{aligned}\int \sin^3(x) \cos^5(x) dx &= \int (\cos^5(x) - \cos^7(x)) \sin(x) dx \\ &= - \int u^5 - u^7 du \\ &= \int u^7 - u^5 du \\ &= \frac{1}{8} u^8 - \frac{1}{6} u^6 + C \\ &= \frac{1}{8} \cos^8(x) - \frac{1}{6} \cos^6(x) + C\end{aligned}$$

Example #4

Find

$$\int \sin^2(x) \cos^2(x) \, dx.$$

Trick:

$$\sin^2(x) = \frac{1 - \cos(2x)}{2} \quad \cos^2(x) = \frac{1 + \cos(2x)}{2}$$

Then

$$\sin^2(x) \cos^2(x) = \frac{1}{4} (1 - \cos^2(2x))$$

$$\cos^2(2x) = \frac{1 + \cos(4x)}{2}$$

so

$$\sin^2(x) \cos^2(x) = \frac{1}{4} \left(1 - \frac{1}{2} - \frac{1}{2} \cos(4x) \right)$$

$$= \frac{1}{4} \left(\frac{1}{2} - \frac{1}{2} \cos(4x) \right)$$

$$\begin{aligned} \int \sin^2(x) \cos^2(x) dx &= \frac{1}{4} \int \frac{1}{2} - \frac{1}{2} \cos(4x) dx \\ &= \frac{1}{4} \left(\frac{1}{2} x - \frac{1}{8} \sin(4x) \right) + C \end{aligned}$$

Idea

$$\int \sin^m(x) \cos^n(x) \, dx$$

1. m odd, use

$$\sin^m(x) = \sin^{m-1}(x) \sin(x) \quad \text{and} \quad \sin^2(x) = 1 - \cos^2(x)$$

2. n odd, use

$$\cos^n(x) = \cos^{n-1}(x) \cos(x) \quad \text{and} \quad \cos^2(x) = 1 - \sin^2(x)$$

3. m and n even, use

$$\sin^2(x) = \frac{1 - \cos(2x)}{2} \quad \text{and} \quad \cos^2(x) = \frac{1 + \cos(2x)}{2}$$

Example #5

Find

$$\int \tan^3(x) \, dx.$$

Trick:

$$\tan^2(x) = \sec^2(x) - 1$$

Here

$$\begin{aligned}\tan^3(x) &= \tan(x)(\sec^2(x) - 1) \\ &= \tan(x)\sec^2(x) - \tan(x)\end{aligned}$$

so

$$\int \tan^3(x) dx = \int \tan(x)\sec^2(x) dx - \int \tan(x) dx$$

Now

$$\int \tan(x)\sec^2(x) dx = \int u du = \frac{1}{2}\tan^2(x) + C_1$$

where

$$u = \tan(x) \quad du = \sec^2(x) dx.$$

While

$$\begin{aligned}\int \tan(x) dx &= \int \frac{\sin(x)}{\cos(x)} dx \\ &= -\int \frac{1}{w} dw \\ &= -\ln|w| + C_2\end{aligned}$$

given

$$w = \cos(x) \quad dw = -\sin(x) dx.$$

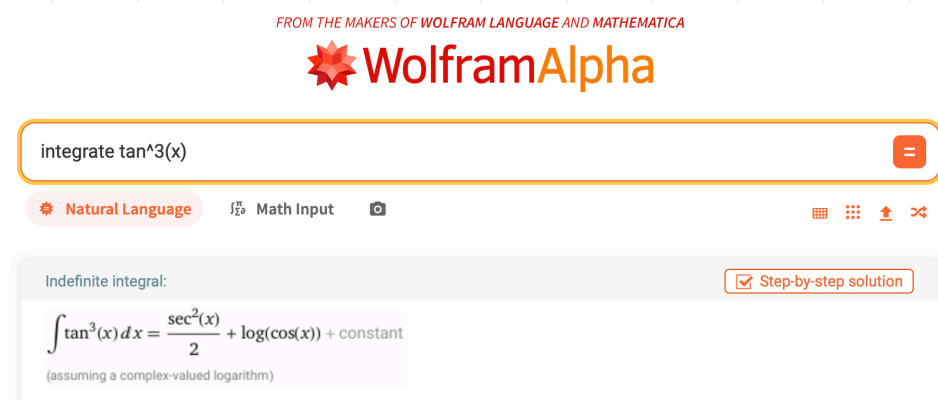
Thus

$$\begin{aligned}\int \tan(x) dx &= -\ln|\cos(x)| + C_2 \\ &= \ln\left|\frac{1}{\cos(x)}\right| + C_2 \\ &= \ln|\sec(x)| + C_2\end{aligned}$$

and so

$$\int \tan^3(x) dx = \frac{1}{2}\tan^2(x) - \ln|\sec(x)| + C$$

Does WolframAlpha give a different result?



How can we rectify this?

Notice that

$$\tan^2(x) = \sec^2(x) - 1$$

so our solutions are equal up to a constant!

FROM THE MAKERS OF WOLFRAM LANGUAGE AND MATHEMATICA



integrate tan^3(x)



 Natural Language

 Math Input



Indefinite integral:

☒ Step-by-step solution

$$\int \tan^3(x) dx = \frac{\sec^2(x)}{2} + \log(\cos(x)) + \text{constant}$$

(assuming a complex-valued logarithm)

Example #6

Find

$$\int \sec^4(x) \, dx.$$

Trick:

$$\sec^2(x) = 1 + \tan^2(x)$$

Notice

$$\begin{aligned}\sec^4(x) &= \sec^2(x) \sec^2(x) \\ &= (1 + \tan^2(x)) \sec^2(x).\end{aligned}$$

$$\int \sec^4(x) dx = \int (1 + \tan^2(x)) \sec^2(x) dx$$

Apply a u-sub

$$u = \tan(x) \quad du = \sec^2(x) dx$$

then

$$\begin{aligned}\int (1 + \tan^2(x)) \sec^2(x) dx &= \int 1 + u^2 du \\ &= u + \frac{1}{3}u^3 + C.\end{aligned}$$

Back substitute to get

$$\int \sec^4(x) dx = \tan(x) + \frac{1}{3}\tan^3(x) + C.$$

Another trick!

$$\sec^4(x) = \frac{1}{\cos^4(x)}.$$

Notice

$$\begin{aligned}\sec^4(x) &= \frac{1}{\cos^4(x)} \\ &= \frac{\cos^2(x) + \sin^2(x)}{\cos^4(x)} \\ &= \left(1 + \frac{\sin^2(x)}{\cos^2(x)}\right) \frac{1}{\cos^2(x)}.\end{aligned}$$

Apply a u-sub

$$u = \frac{\sin(x)}{\cos(x)} \quad du = \frac{1}{\cos^2(x)} dx$$

then

$$\begin{aligned}\int \frac{1}{\cos^4(x)} dx &= \int 1 + u^2 du \\ &= u + \frac{1}{3}u^3 + C.\end{aligned}$$

Back substitute to get

$$\int \sec^4(x) dx = \tan(x) + \frac{1}{3}\tan^3(x) + C.$$

Idea

$$\int \tan^m(x) \sec^n(x) \, dx$$

1. m odd, use

$$\tan^2(x) = \sec^2(x) - 1 \quad \text{and} \quad \text{“save” a } \sec(x) \tan(x)$$

2. n even, use

$$\sec^2(x) = 1 + \tan^2(x) \quad \text{and} \quad \text{“save” a } \sec^2(x)$$

3. otherwise, experiment...

Example #7

Find

$$\int \cot(x) \, dx.$$

With

$$\cot(x) = \frac{\cos(x)}{\sin(x)}$$

let

$$u = \sin(x) \quad du = \cos(x) \, dx.$$

Then

$$\begin{aligned} \int \cot(x) \, dx &= \int \frac{\cos(x)}{\sin(x)} \, dx \\ &= \int \frac{1}{u} \, du \\ &= \ln|\sin(x)| + C. \end{aligned}$$